Vasicek vs CR+

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1 Introduction

The Vasicek model and CreditRisk+ (CR+) are two established models for credit portfolio risk. Despite significant differences in mathematical structure, the models in practice describe the same systematic credit risk in large homogeneous portfolios. The difference lies mainly in how this risk is represented:

- The Vasicek model uses a latent normally distributed systematic factor with asset correlation ρ.
- CreditRisk+ uses a stochastic default intensity Λ which is assumed to be gamma distributed.

Both the Vasicek model and CreditRisk+ belong to the so-called ASRF class (*Asymptotic Single Risk Factor*) of portfolio models. This type of model is formulated such that the model contains a limiting case, *the large portfolio limit*, where idiosyncratic risk is diversified away and credit losses are entirely driven by a common systematic risk factor.[1]

In the large portfolio limit, the models can be calibrated such that they produce almost identical loss distributions. This can be done either by matching the variance of the common risk driver (the standard method), or by matching a certain quantile in the loss distribution, for example 99 %-VaR. Both methods are discussed below.

2 Common Risk Structure

2.1 Vasicek

In the Vasicek model, each borrower is assumed to default if a latent variable falls below a threshold:

$$X = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon,$$

[1]ASRF (*Asymptotic Single Risk Factor*) is a portfolio framework for credit risk where losses are assumed to be driven by a single common systematic risk factor in the limit of a very large and well-diversified portfolio. In this limit – *the large portfolio limit* – idiosyncratic risk is diversified away and the loss distribution is entirely determined by the variations in the common risk driver. The framework constitutes the theoretical foundation for the Vasicek model as well as the Basel regulatory internal ratings-based (IRB) approach for credit risk.

where $Z \sim N(0,1)$ is a common systematic factor and $\varepsilon \sim N(0,1)$ is idiosyncratic risk. Default occurs if

$$X < \Phi^{-1}(p),$$

which gives the conditional probability of default

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}Z}{\sqrt{1-\rho}} \right).$$

All systematic credit risk thus operates through the fact that the *conditional* PD is stochastic.

2.2 CreditRisk+

In CreditRisk+, the number of defaults is modeled as Poisson distributed given an intensity:

$$N \mid \Lambda \sim \text{Poisson}(\Lambda).$$

Systematic dependence is introduced by making the intensity itself stochastic:

$$\Lambda \sim \text{Gamma}.$$

All exposures share the same intensity, which creates correlated default outcomes. In CreditRisk+, the gamma distribution is used to model variation in a common default intensity. The gamma distribution is a continuous distribution which is defined for positive values and is suitable for describing skewed variation around a given mean value. The gamma distribution can be parameterized with a shape parameter α and a scale parameter β. For a gamma-distributed stochastic variable Λ it holds that

$$E[\Lambda] = \alpha\beta, \qquad \text{Var}(\Lambda) = \alpha\beta^2.$$

The shape parameter thus controls the relative dispersion, while the scale parameter determines the level scale.

A central measure in credit risk context is the so-called gamma variance, defined as the relative variance in the default intensity:

$$\text{CV}^2 = \frac{\text{Var}(\Lambda)}{E[\Lambda]^2}.$$

For a gamma distribution, this expression reduces to

$$\text{CV}^2 = \frac{\alpha\beta^2}{(\alpha\beta)^2} = \frac{1}{\alpha}.$$

which shows that the gamma variance depends solely on the shape parameter and constitutes a dimensionless measure of the degree of systematic variation in the portfolio's common risk driver.

The gamma variance affects the extent to which default events cluster in adverse system states and thus has a decisive impact on the tail of the loss distribution and economic capital. The expected loss, however, is not affected, as it is determined by the mean value of the intensity.

2.3 Common Denominator

In large homogeneous portfolios, idiosyncratic risk is diversified away (that is, *in the large portfolio limit*). The loss distribution is then entirely determined by variation in the common risk driver:

- conditional PD in the Vasicek model,
- intensity in CreditRisk+.

This is particularly decisive for the tail of the loss distribution.

2.4 Example: how systematic dependence arises in CreditRisk+

To concretely illustrate the mechanism in CreditRisk+, consider a simple, homogeneous credit portfolio with many small exposures.

Assume that the portfolio consists of 10 000 identical loans with unconditional probability of default of 1 %. The expected number of defaults is then

$$E[\Lambda] = 10\,000 \times 0.01 = 100.$$

In CreditRisk+, the number of defaults N given the common default intensity Λ is modeled as Poisson distributed:

$$N \mid \Lambda \sim \text{Poisson}(\Lambda).$$

The decisive assumption is that the intensity itself is stochastic:

$$\Lambda \sim \text{Gamma}.$$

System states and clustering of defaults

The interpretation is that each observation period (e.g. one year) realizes a system state that determines the value of Λ. Given this value, defaults occur in the portfolio otherwise independently.

To illustrate this, one can consider three different system states:

- A very favorable year, where $\Lambda \approx 60$, which gives an unusually low number of defaults.
- An average year, where $\Lambda \approx 100$, which gives approximately 100 defaults.
- A very unfavorable year, where $\Lambda \approx 160$, which gives a large cluster of defaults.

Even though default outcomes are independent given Λ, the variation in Λ leads to many loans defaulting simultaneously in unfavorable system states. This creates positive correlation between default outcomes for different exposures.

Economic interpretation

It is this variation in Λ that corresponds to the variation in conditional probability of default that arises in the Vasicek model through the systematic factor. CreditRisk+ thus achieves correlated default outcomes without modeling any latent factor explicitly; all system risk is concentrated in the stochastic intensity.

2.5 Parallel example: how systematic dependence arises in the Vasicek model

To illustrate the parallel with CreditRisk+, we now consider the same portfolio in the Vasicek model.

Assume again a portfolio with 10 000 identical loans with unconditional probability of default of 1 %. The expected loss thus corresponds on average to 100 defaults per year.

In the Vasicek model, each borrower is assumed to default if

$$X_i = \sqrt{\rho}\, Z + \sqrt{1-\rho}\, \varepsilon_i < \Phi^{-1}(p),$$

where $Z \sim N(0,1)$ is a common systematic factor and $\varepsilon_i \sim N(0,1)$ is idiosyncratic risk.

System states and conditional PD

Given a realized value of the systematic factor $Z = z$, the probability of default becomes the same for all loans and is given by

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, z}{\sqrt{1-\rho}} \right).$$

The outcome in the portfolio can then be interpreted as a system state determined by z. To provide intuition, one can again consider three representative states:

- A favorable state with high z, where $p(z)$ is significantly lower than 1 % and very few loans default.
- An average state around $z \approx 0$, where $p(z) \approx 1$ % and the number of defaults is close to 100.
- An adverse state with low z, where $p(z)$ becomes strongly elevated and a large cluster of loans defaults.

Given $Z = z$, default outcomes in the portfolio are otherwise independent. Just as in CreditRisk+, it is therefore the variation in the common system state, not idiosyncratic randomness, that drives clustering of default.

Economic interpretation

In the Vasicek model, correlated default outcomes thus arise because all loans are affected simultaneously by the systematic factor Z. When Z takes adverse values, the conditional PD for the entire portfolio increases, which leads to a large number of simultaneous defaults.

The mechanism is thus directly analogous to CreditRisk+, where the same type of clustering arises through a high realized value of the intensity Λ. The difference between the models lies in the representation, not in the underlying economic risk.

3 Derivation of the relationship between ρ and gamma variance

The central relationship between the asset correlation ρ in the Vasicek model and the gamma volatility in CreditRisk+ can be derived by studying the variation in the common risk driver in the large portfolio limit.

3.1 Conditional PD as a random variable in the Vasicek model

In the Vasicek model, the conditional probability of default is given by

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, Z}{\sqrt{1-\rho}}\right),$$

where $Z \sim N(0,1)$ is the systematic factor.

Since Z is random, $p(Z)$ is also a random variable. For low unconditional PD and moderate variations in Z, the function $p(Z)$ can be linearized around $Z = 0$:

$$p(Z) \approx p(0) + p'(0)\, Z.$$

The derivative is given by

$$p'(Z) = -\frac{\sqrt{\rho}}{\sqrt{1-\rho}}\, \varphi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, Z}{\sqrt{1-\rho}}\right),$$

where φ is the density function of the standard normal distribution. Evaluated at $Z = 0$, we obtain

$$p'(0) = -\frac{\sqrt{\rho}}{\sqrt{1-\rho}}\, \varphi\left(\frac{\Phi^{-1}(p)}{\sqrt{1-\rho}}\right).$$

3.2 Approximation for low PD

For low values of p, the approximation holds[2]

$$\varphi\left(\frac{\Phi^{-1}(p)}{\sqrt{1-\rho}}\right) \approx \sqrt{1-\rho}\,\varphi\big(\Phi^{-1}(p)\big)\,.$$

Inserted into the expression above gives

$$p'(0) \approx -\sqrt{\rho}\,\varphi\big(\Phi^{-1}(p)\big)\,.$$

The variance of the conditional PD can thus be approximated as

$$\mathrm{Var}[p(Z)] \approx (p'(0))^2 \mathrm{Var}(Z) = \rho\,\varphi\big(\Phi^{-1}(p)\big)^2\,.$$

Dividing this by the square of the unconditional PD gives the relative variance:

$$\frac{\mathrm{Var}[p(Z)]}{E[p(Z)]^2} \approx \rho\left(\frac{\varphi(\Phi^{-1}(p))}{p}\right)^2.$$

For low PD, the ratio $\varphi(\Phi^{-1}(p))/p$ grows at the same order as $(1-\rho)^{-1/2}$, which leads to the ASRF-established result

$$\frac{\mathrm{Var}[p(Z)]}{E[p(Z)]^2} \approx \frac{\rho}{1-\rho}.$$

3.3 Relative variance in CreditRisk+

In CreditRisk+, the default intensity Λ is assumed to be gamma distributed. For a gamma distribution, it holds exactly that

$$\frac{\mathrm{Var}(\Lambda)}{E[\Lambda]^2} = \mathrm{CV}^2,$$

where CV^2 is the squared coefficient of variation, which is the gamma volatility.

[2] In Vasicek and ASRF-based portfolio models, terms of the type $\varphi(c/\sqrt{1-\rho})$ appear, where $c = \Phi^{-1}(p)$ is the default threshold and ρ denotes the degree of systematic dependence. For low probabilities of default, $c \to -\infty$, which implies that asymptotic tail approximations for the normal distribution can be applied. In this regime, the approximation

$$\varphi\left(\frac{c}{\sqrt{1-\rho}}\right) \approx \sqrt{1-\rho}\,\varphi(c),$$

is often used. It is not exact but valid asymptotically for low default levels. The approximation is used to obtain analytically tractable expressions for the variation in conditional credit risk and underlies central results in the ASRF framework, such as the relationship between asset correlation and relative variance in conditional probability of default.

3.4 Identification in the large portfolio limit

In the large portfolio limit, the loss distribution is entirely driven by variation in the common risk driver. For the Vasicek model and CreditRisk+ to produce the same loss distribution, the relative variance of this risk driver must therefore be the same.

By identifying the stochastic PD in the Vasicek model with the intensity in CreditRisk+, we obtain

$$\mathrm{CV}^2 \approx \frac{\rho}{1-\rho}.$$

Solving with respect to ρ yields the commonly used relationship

$$\rho \approx \frac{\mathrm{CV}^2}{1+\mathrm{CV}^2}.$$

This shows that the gamma volatility in CreditRisk+ is an (approximately) alternative parameterization of the same systematic credit risk as the asset correlation in the Vasicek model.

4 Calibration of correlation and gamma volatility

Assume that the asset correlation in Vasicek is $\rho = 10\,\%$. At a 99 % confidence level, the outcome corresponds to a strongly negative realization of the systematic factor Z. The conditional PD then becomes significantly higher than 1 %, which leads to a large cluster of defaults.

4.1 Problem formulation

By exploiting the relationship above, we can translate this into CR+, where the starting point is the relationship between the asset correlation ρ in the Vasicek model and the squared coefficient of variation (gamma volatility) in CreditRisk+:

$$\mathrm{CV}^2 \approx \frac{\rho}{1-\rho}.$$

In the numerical example, the asset correlation is assumed to be

$$\rho = 10\,\% = 0.10.$$

Inserted into the expression above, this yields

$$\mathrm{CV}^2 \approx \frac{0.10}{1-0.10} = \frac{0.10}{0.90} \approx 0.111.$$

By definition, the coefficient of variation satisfies

$$\mathrm{CV} = \frac{\sqrt{\mathrm{Var}(\Lambda)}}{E[\Lambda]}.$$

4.2 Solution for CV

The standard deviation relative to the mean is therefore obtained as

$$\mathrm{CV} = \sqrt{\mathrm{CV}^2} = \sqrt{0.111} \approx 0.333.$$

4.3 Economic interpretation

This implies that the standard deviation of the common default intensity Λ is approximately 33 % of its mean. At the 99 % quantile, a high value of Λ is realized, which yields essentially the same number of defaults as in the Vasicek case. Numerical comparisons can show that VaR_{99} differs very little between the two models.

5 Calibration of ρ to a given loss level (e.g. 99%)

In practical applications, it may be desirable not only to match the variance of the common risk driver, but to match a specific quantile of the loss distribution. This is particularly relevant if CR+ is used as the primary model and one wishes to find a corresponding Vasicek correlation.

5.1 Problem formulation

Assume that CR+ produces a 99% loss level (VaR) of

$$L_{0.99}^{\mathrm{CR+}} = \ell_{99}.$$

We then seek the correlation ρ in the Vasicek model that produces the same quantile:

$$L_{0.99}^{\mathrm{Vasicek}}(\rho) = \ell_{99}.$$

5.2 Large portfolio limit

In a homogeneous portfolio with LGD = 1 and exposures normalized to 1, the loss in the Vasicek model in the large portfolio limit is

$$L(z) = p(z),$$

where z is the realization of the systematic factor. The 99% loss is then given by

$$L_{0.99}^{\mathrm{Vasicek}} = p(z_{0.99}), \qquad z_{0.99} = \Phi^{-1}(0.99).$$

Inserted into the expression for $p(z)$, we obtain

$$p(z_{0.99}) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right).$$

5.3 Solution for ρ

We want to solve the equation

$$\Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right) = \ell_{99}.$$

This can be rewritten as

$$\frac{\Phi^{-1}(p) - \sqrt{\rho}\,\Phi^{-1}(0.99)}{\sqrt{1-\rho}} = \Phi^{-1}(\ell_{99}).$$

The solution for ρ becomes

$$\sqrt{\rho} = \frac{\Phi^{-1}(p) - \Phi^{-1}(\ell_{99})\sqrt{1-\rho}}{\Phi^{-1}(0.99)}.$$

The equation is analytically solvable, but it is easiest to solve it numerically. Since it is monotonic in ρ, a unique solution is guaranteed.

5.4 Economic interpretation

This calibration means that one chooses ρ such that the Vasicek model reproduces the same extreme loss as CR+. This is particularly useful if CR+ is used for capital calculation, but one wishes to express the corresponding systematic risk in terms of a Vasicek correlation.

The method is also robust: even though the models differ in tail behavior, calibration to a specific quantile yields a directly comparable risk level.

6 Conclusion

The Vasicek model and CreditRisk+ are two mathematical representations of the same underlying systematic risk. With correct translation between asset correlation and gamma volatility, the models in large homogeneous portfolios produce almost identical risk measures, especially at high confidence levels such as 99% and above.